

Mount Kenya University

Unlocking Infinite Possibilities

UNIT TITLE: ADVANCED WEB

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ADMISSION NUMBER: BSCCS/2024/55278

COURSE: B.S.C COMPUTER SCIENCE

LECTURE: MR NYORO MICHAEL

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GitHub repo: <https://github.com/LuizSamuel/advanced-web-barber>

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WEEK 1: LOCAL ENVIRONMENT SETUP

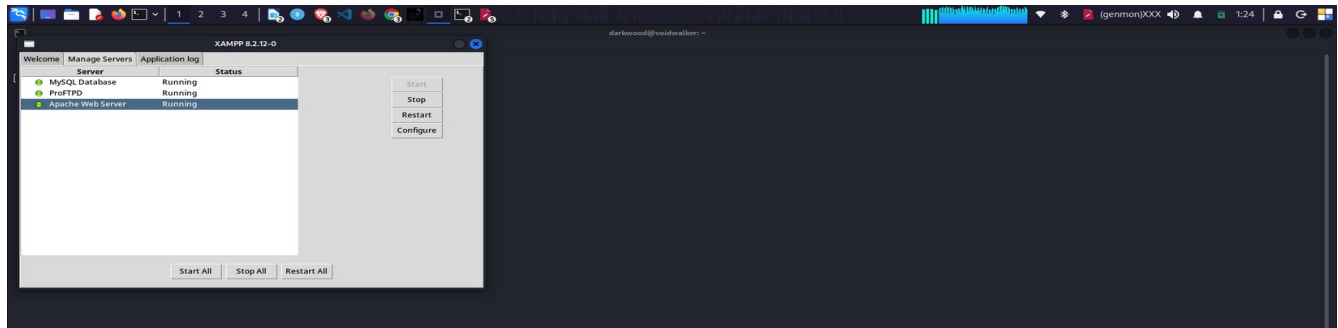


Figure 1: Apache and MySQL Running

This figure confirms that the local development environment was successfully configured using the XAMPP control panel

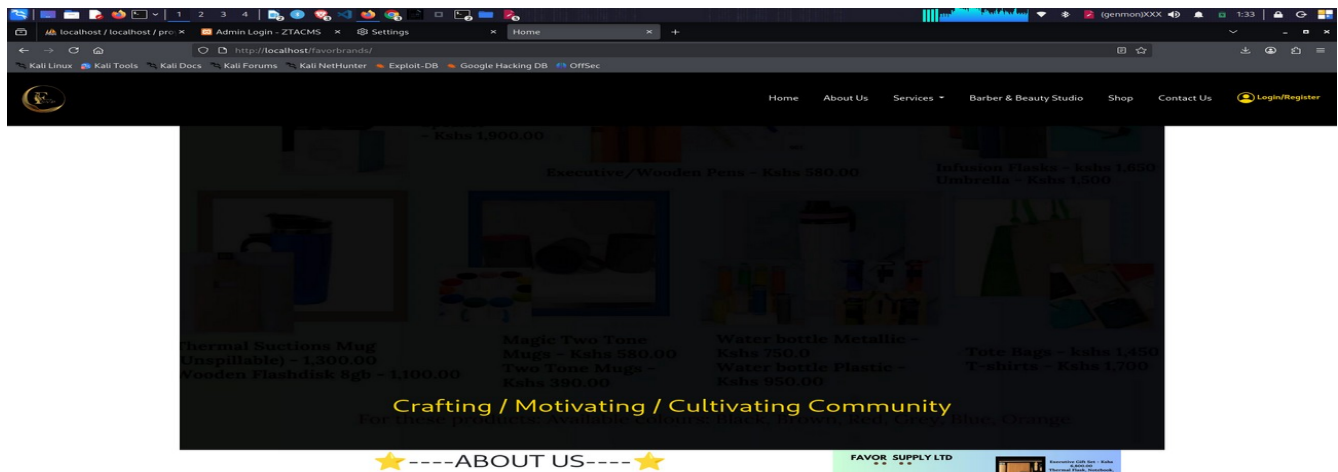


Figure 2: Localhost Test Page

This figure verifies that the local loopback address resolves correctly in the browser, confirming that the development server is actively receiving and responding to HTTP requests as expected.

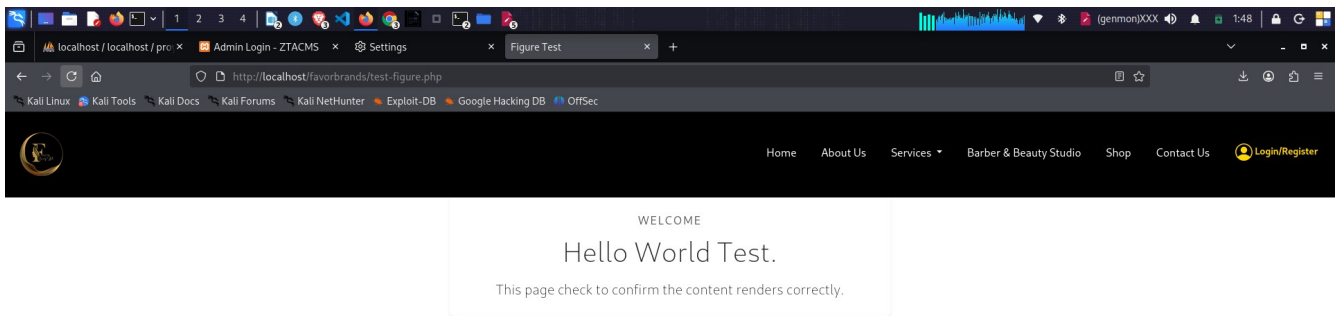


Figure 3: Hello World Test

This figure demonstrates a basic PHP script execution on the local server, serving as an initial verification that the PHP engine is correctly installed,



Figure 4: Database Connection Test

This figure confirms a successful connection between the PHP application layer and the local MySQL database instance

WEEK 2: WIREFRAMES AND GUI DESIGN

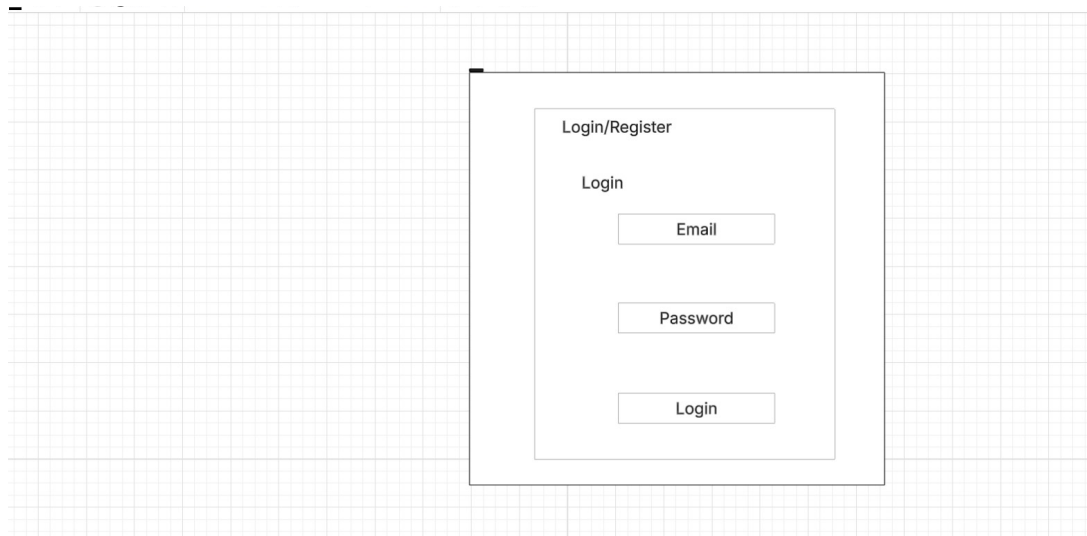


Figure 1: Hand-Drawn Wireframe

This figure presents the initial conceptual layout of the web application interface, capturing early design decisions regarding page structure, content hierarchy, and element placement

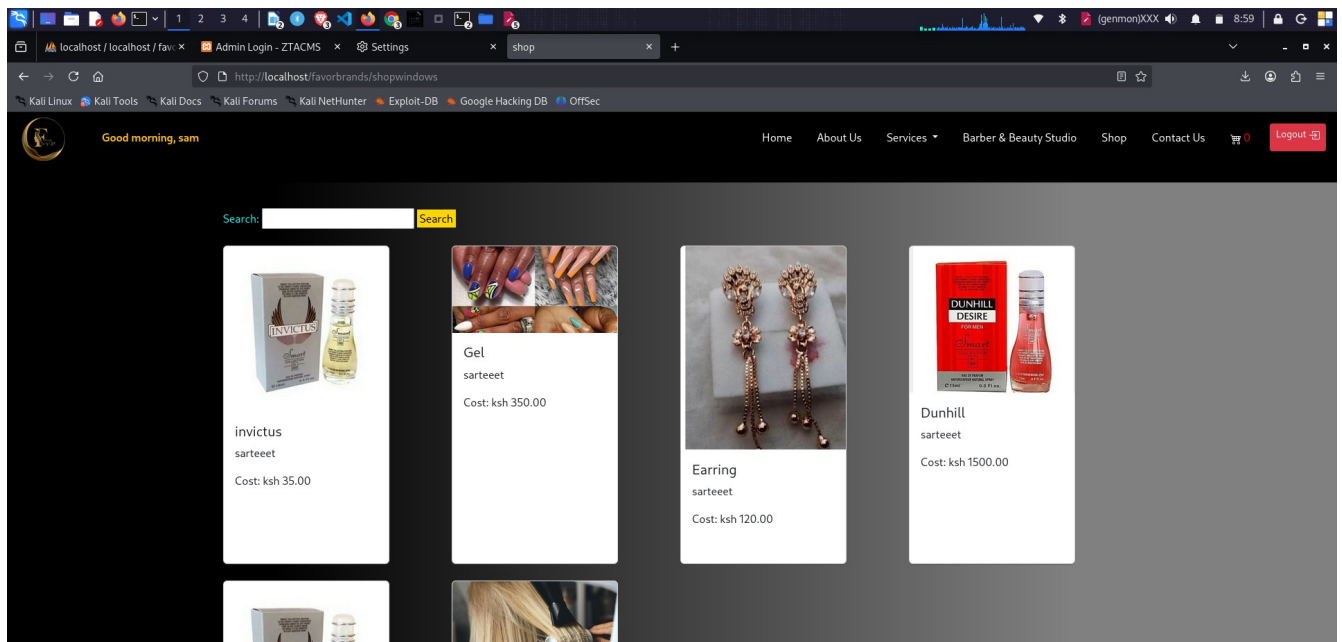


Figure 2: Dashboard Layout Design

This figure illustrates the structured draft of the user interface, defining the positioning of navigation controls, content panels, and key interactive elements across the main dashboard view.

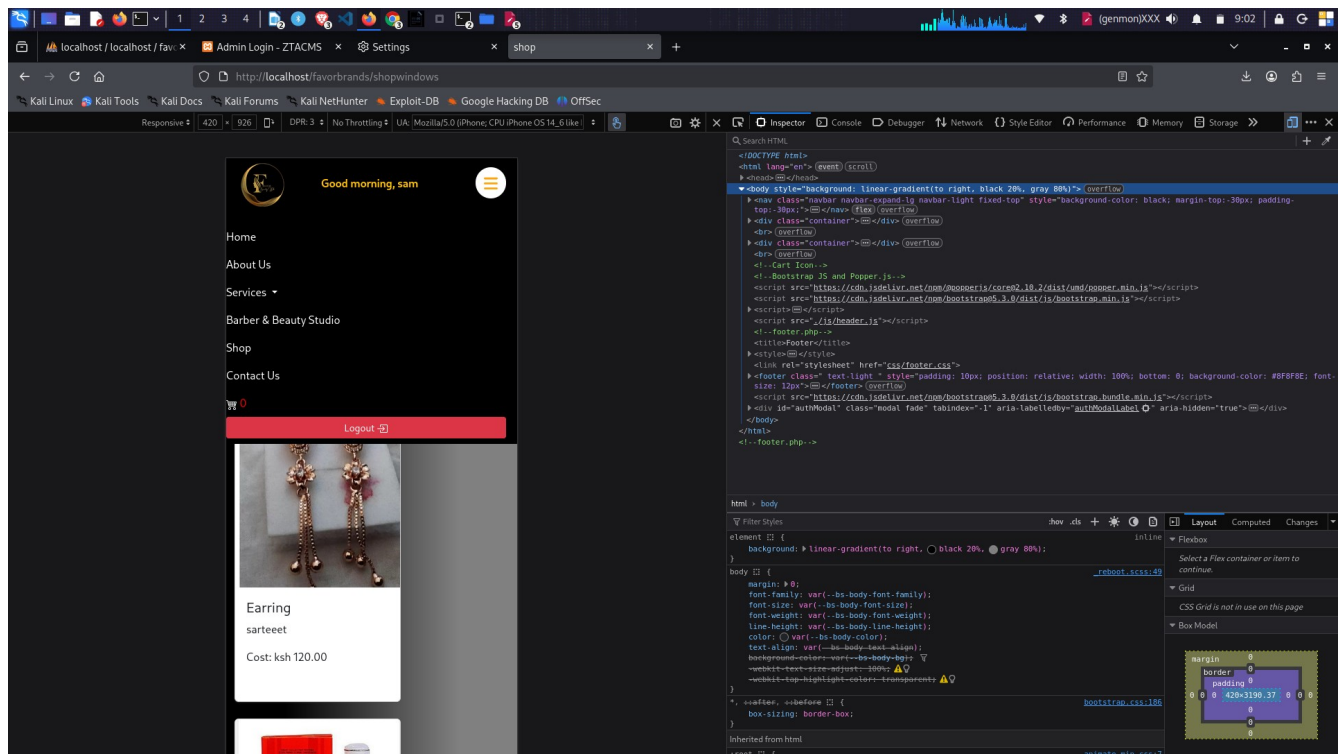


Figure 3: Mobile View Mockup

This figure demonstrates the responsive design approach adopted for the application, showing how navigation menus and image assets adapt and reflow to fit smaller mobile device screen.

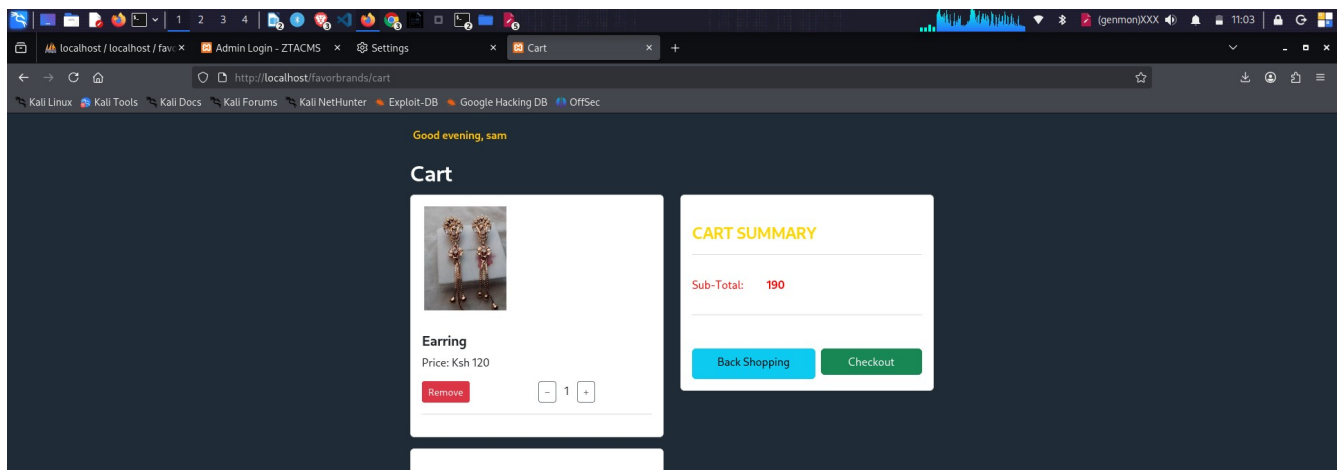


Figure 4: Colour Theme and Navigation Structure

This figure illustrates the visual design language established for the system, intuitive user experience across the application.

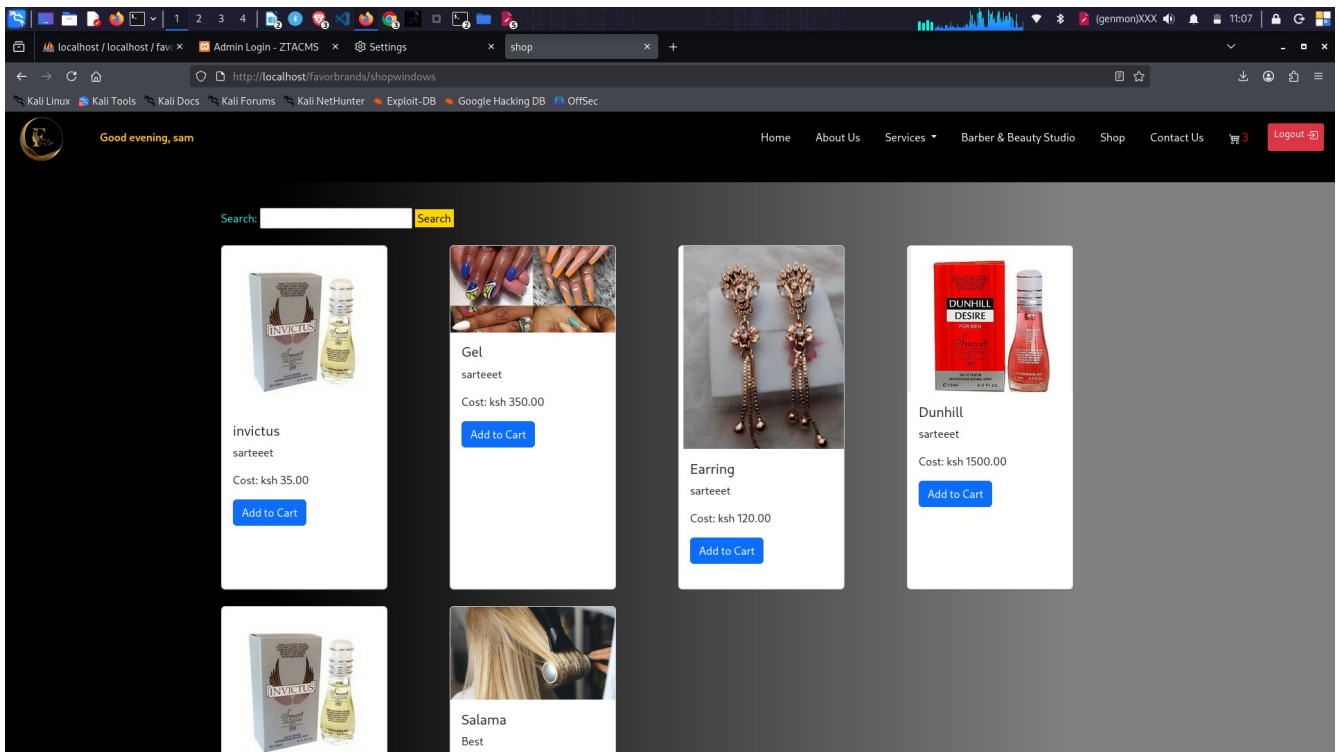


Figure 5: Displaying styled listing cards

with associated pricing and technical specifications, providing a realistic preview of the production interface.

WEEK 3: SCRIPTING AND BACKEND FUNDAMENTALS

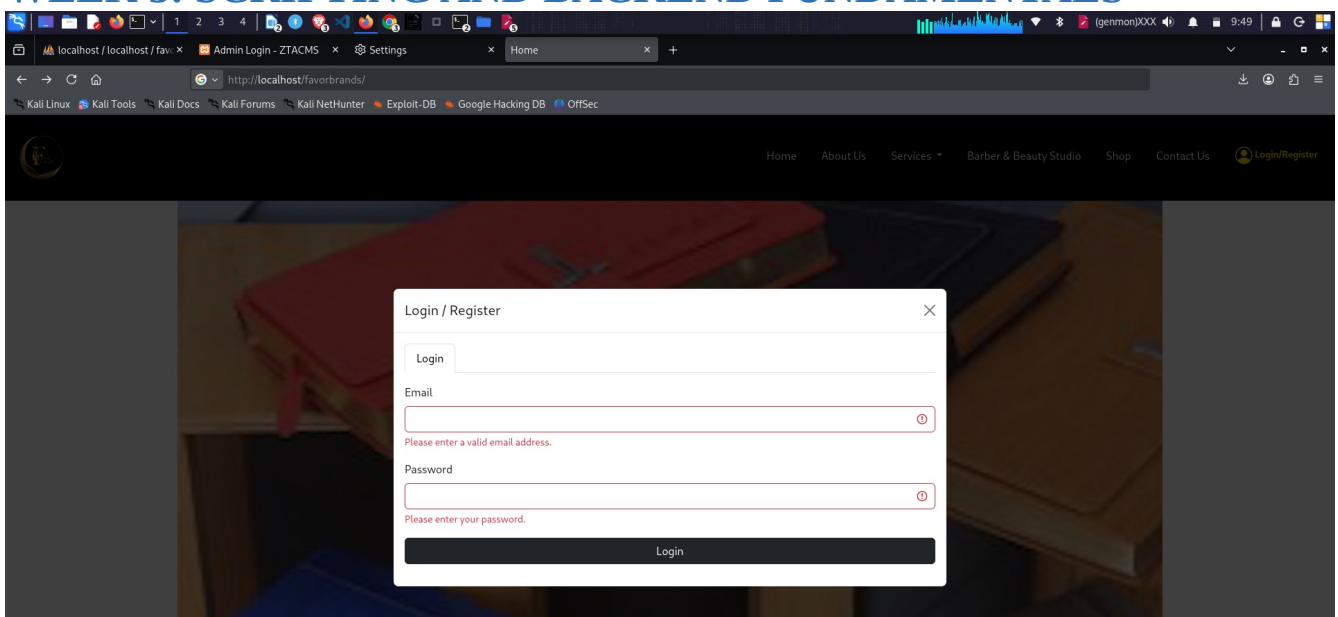


Figure 1: JavaScript Form Validation

This figure demonstrates client-side input validation logic

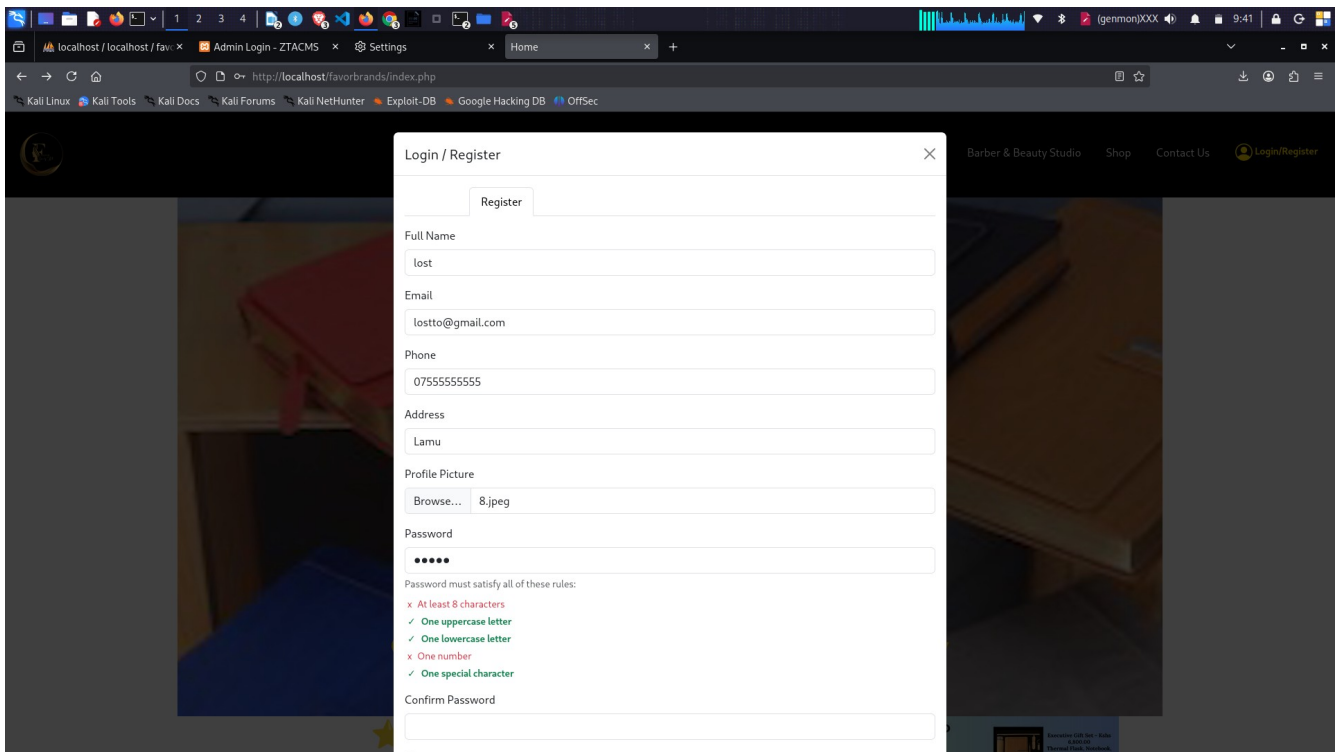


Figure 2: Password Strength Checker

This assess password entropy by evaluating a combination of character length, uppercase and lowercase letters, numeric digits, and special characters, guiding users toward stronger credential choices.

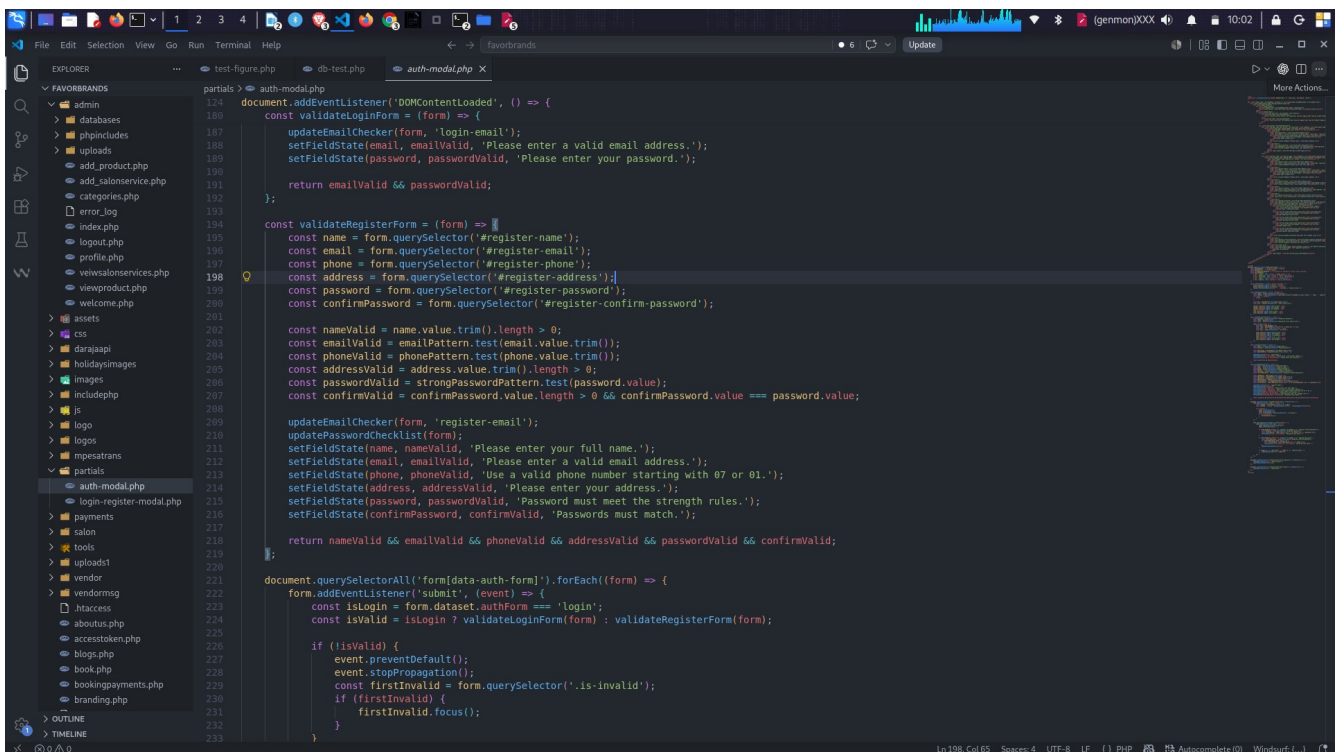


Figure 3: PHP Syntax Practice

This figure illustrates a server-side validation script that captures form state, handles dynamic error message rendering, and toggles CSS error classes conditionally based on the validity of each input field submitted.

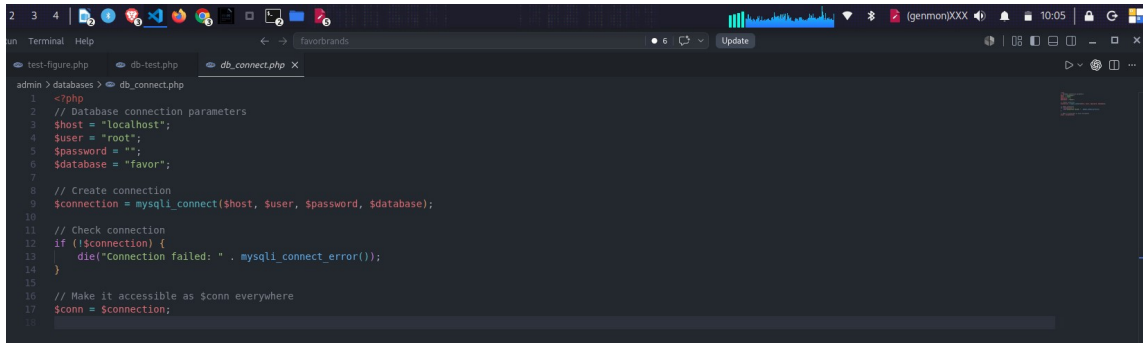


Figure 4: Database Connection Script

This figure shows a backend configuration script that establishes a secure PHP Data Objects (PDO) connection to the database

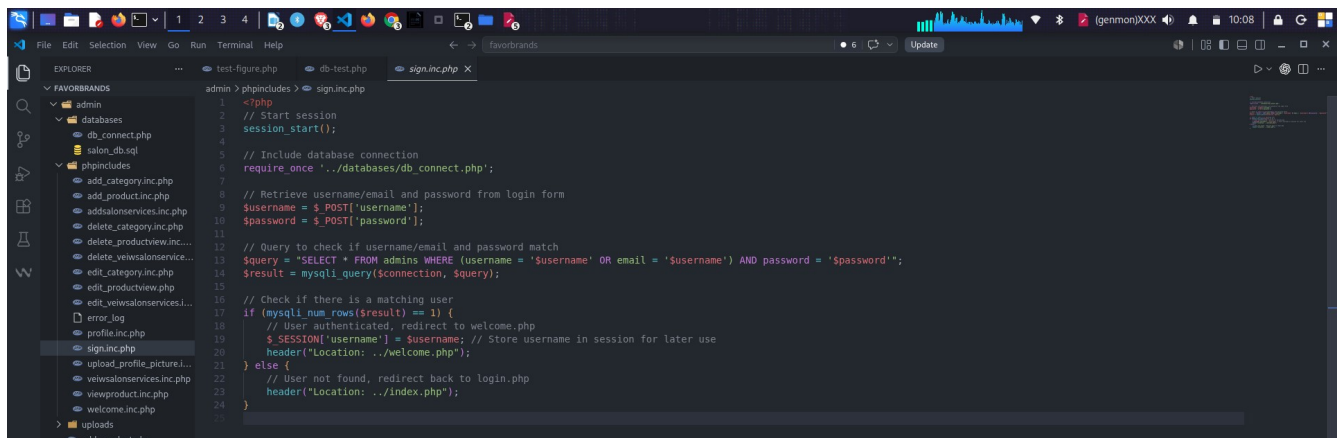


Figure 5: Dynamic User Input Handling

This figure demonstrates PHP server-side code that captures POST request parameters including username, email, and password

WEEK 4: AUTHENTICATION INTERFACE AND PROJECT STRUCTURE

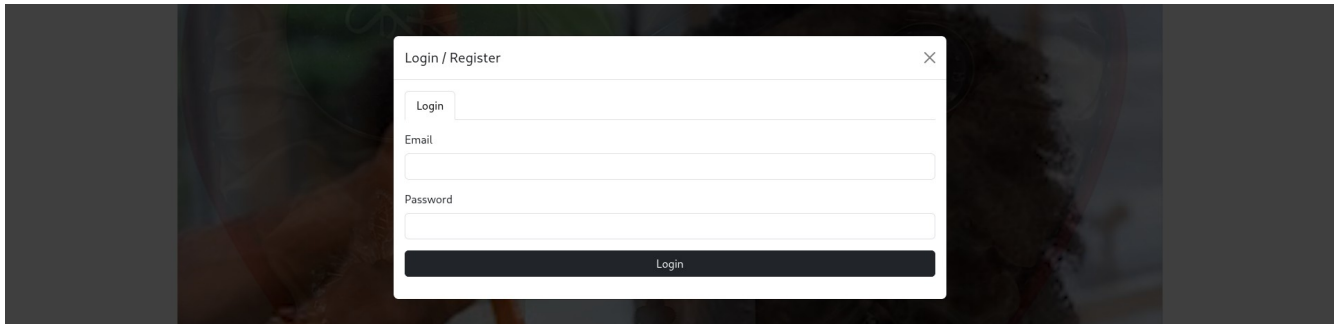


Figure 1: Login Form

This figure presents the user-facing authentication interface

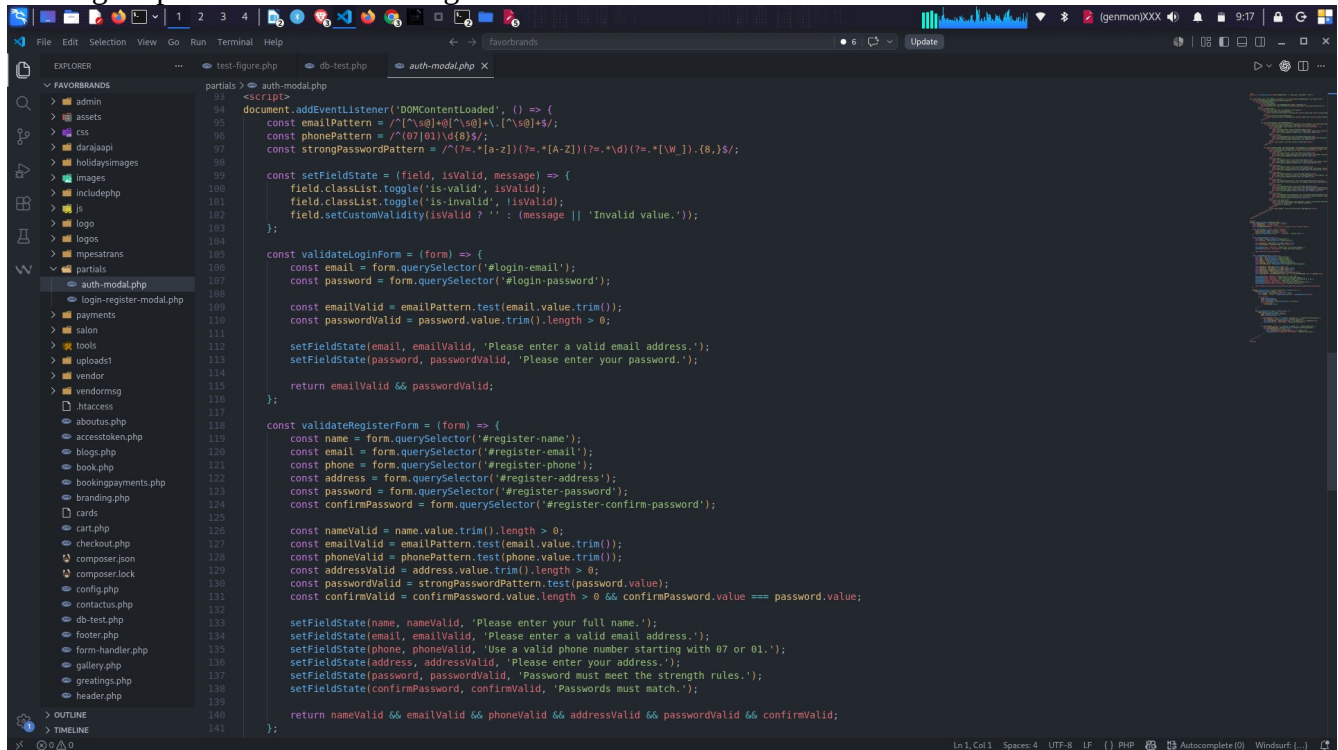


Figure 2: JavaScript Form Validation on Login

This figure demonstrates the integration of client-side validation logic directly into the login and registration forms

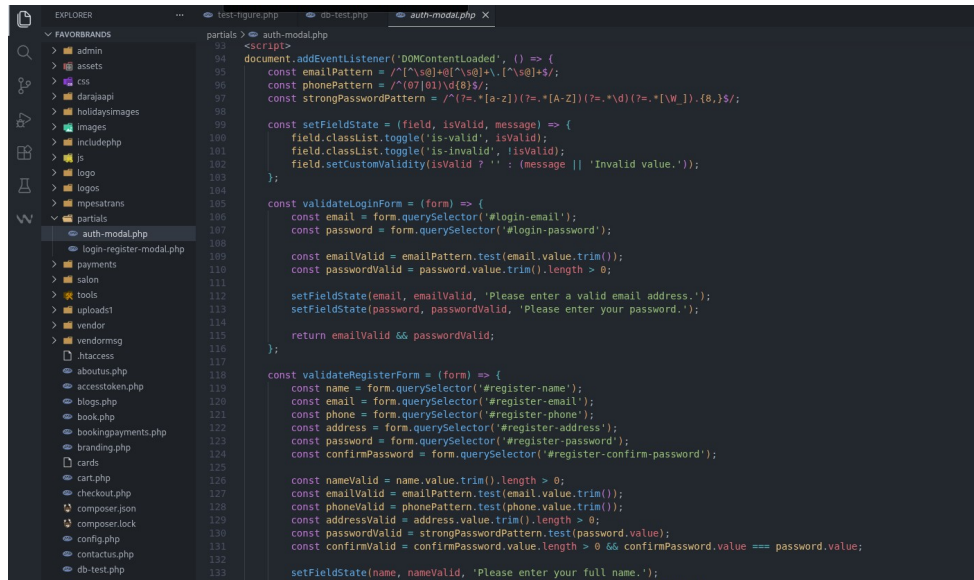


Figure 3: Folder Structure

This figure illustrates the organised project directory structure

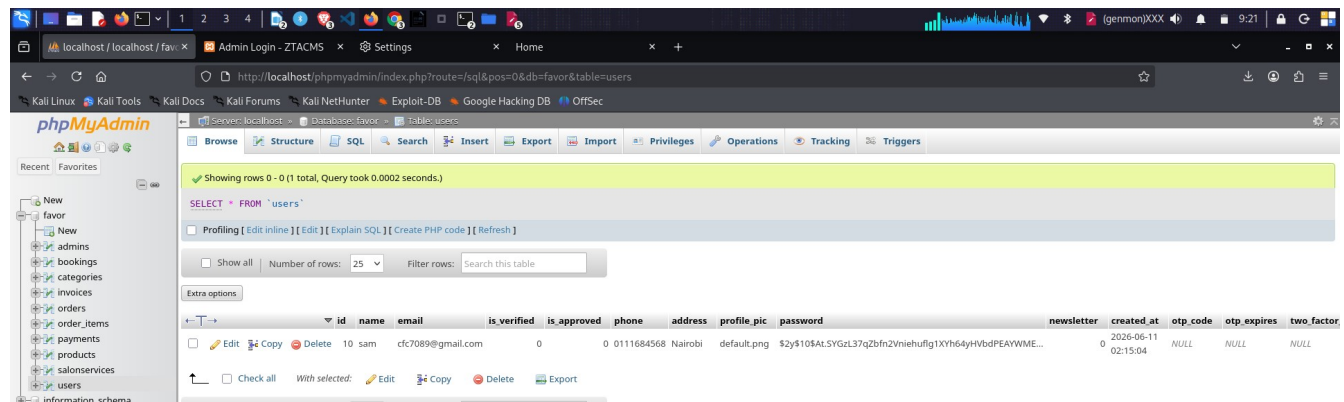


Figure 4: Database Connection and Users Table

This figure confirms the physical deployment structure within the MySQL database

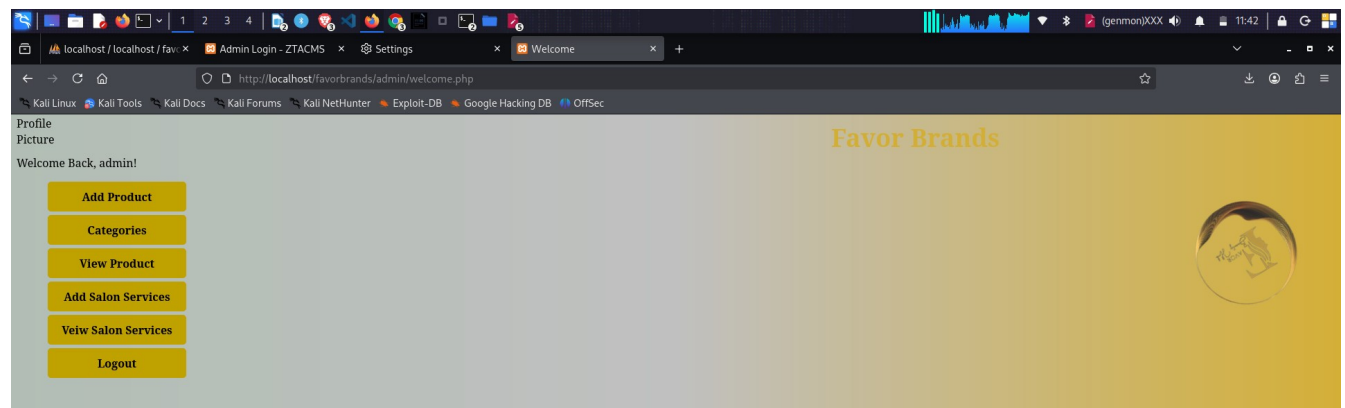


Figure 5: The admin panel

WEEK 5: DATABASE OPERATIONS AND CRUD FUNCTIONALITY

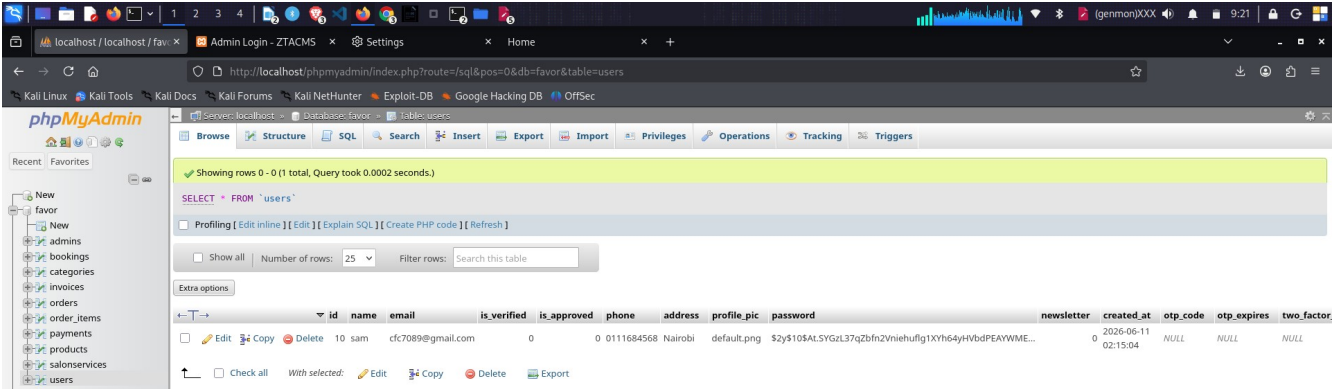


Figure 1: Users Table Structure

This figure presents the schema of the users table within the MySQL database

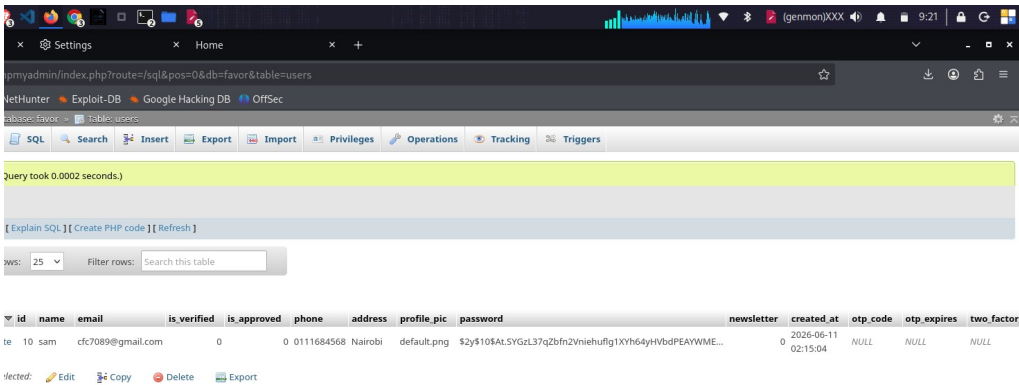


Figure 2: User Email and Password Record

This figure demonstrates the successful insertion and retrieval of a user record from the users table

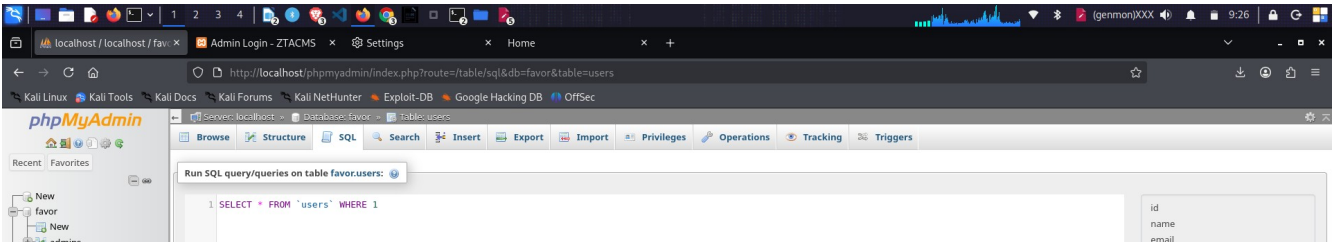
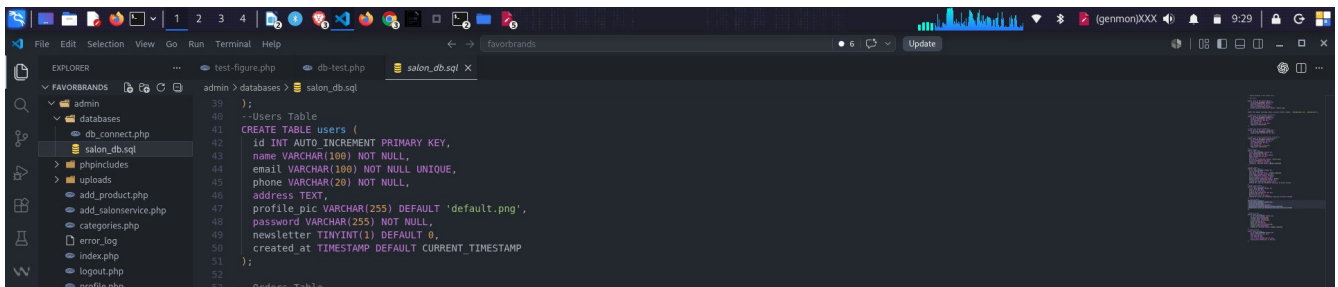


Figure 3: Display All Users Query

This figure shows the execution and output of a SQL SELECT statement against the users table, retrieving and rendering all records currently stored within the authentication table to verify data integrity and insertion accuracy.



```
39 );
40 --Users Table
41 CREATE TABLE users (
42   id INT AUTO_INCREMENT PRIMARY KEY,
43   name VARCHAR(100) NOT NULL,
44   email VARCHAR(100) NOT NULL UNIQUE,
45   phone VARCHAR(20) NOT NULL,
46   address TEXT,
47   profile_pic VARCHAR(255) DEFAULT 'default.png',
48   password VARCHAR(255) NOT NULL,
49   newsletter TINYINT(1) DEFAULT 0,
50   created_at TIMESTAMP DEFAULT CURRENT_TIMESTAMP
51 );
52 --Users Table
```

Figure 4: Users Table Creation Script

This figure presents the SQL Data Definition Language script used to programmatically define the users table structure

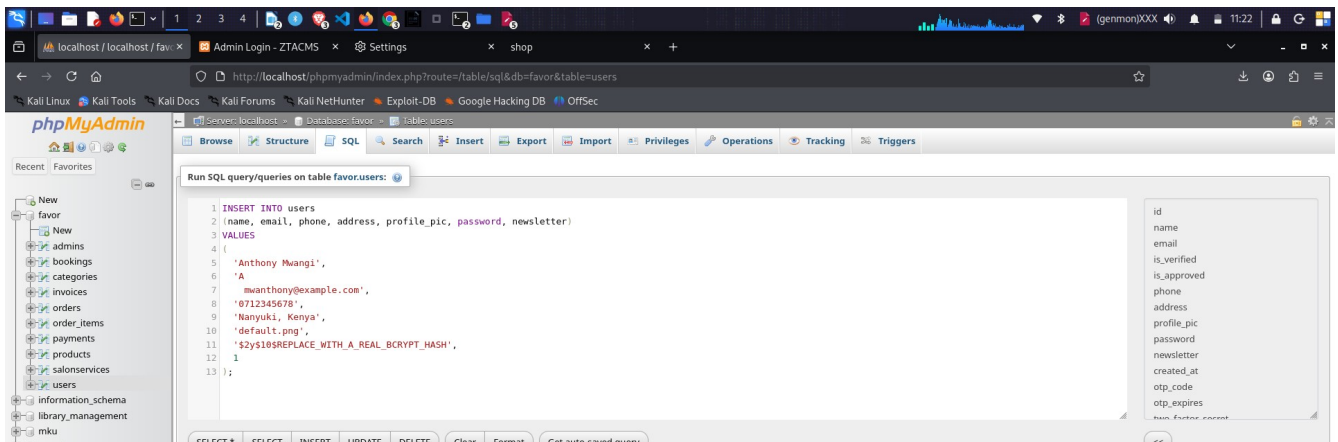


Figure 5: Record Insertion into Table

This figure illustrates the execution of a Data Manipulation Language INSERT statement

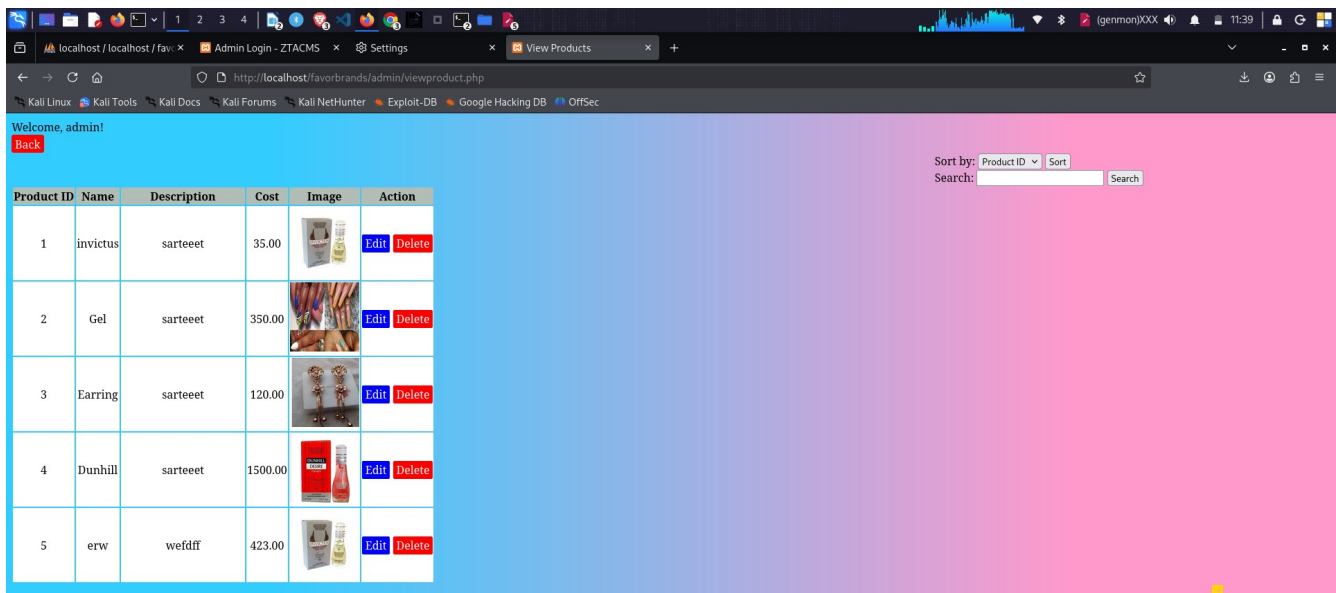


Figure 6: Product CRUD Page

This figure demonstrates a functional inventory management interface featuring a structured data table with full Create, Read, Update, and Delete capabilities, allowing authorised users to manage product entries including item names, prices, and stock levels.

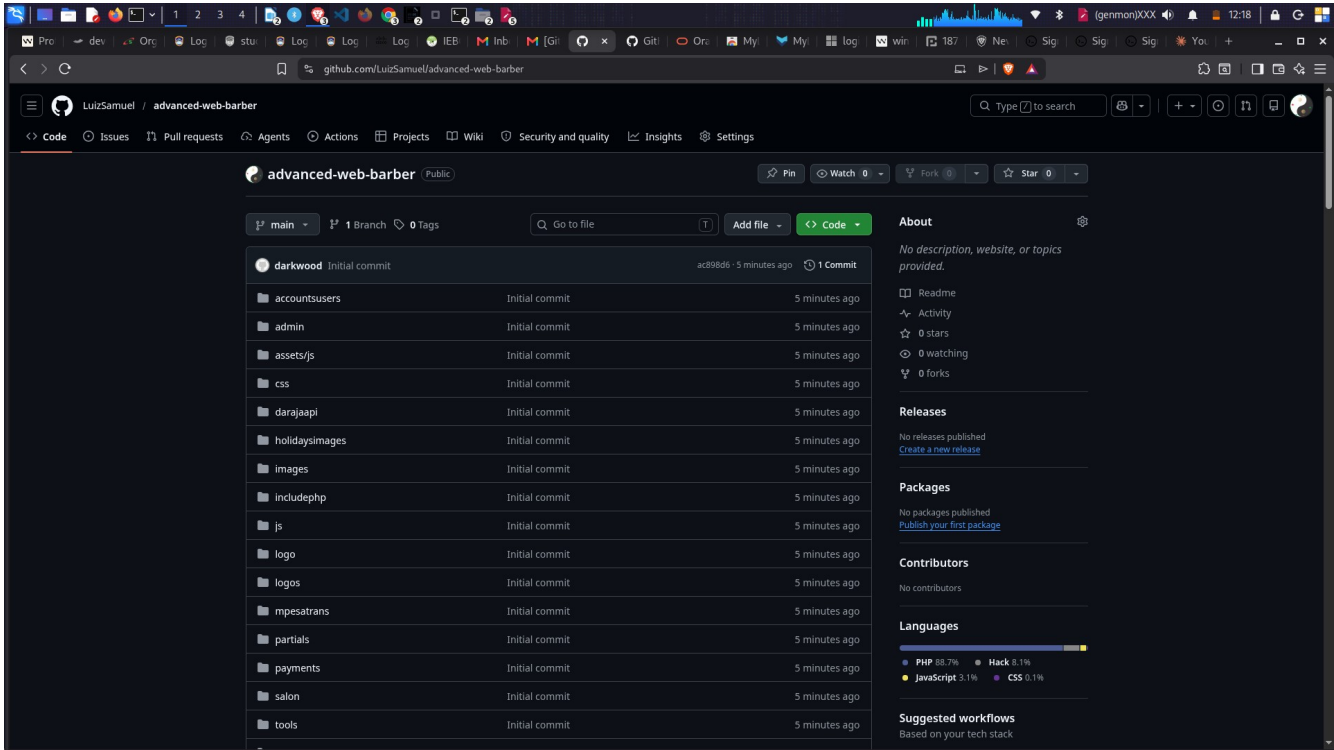


Figure 7: GitHub Weekly Update

This figure captures the version control history for Week 5, showing a modular repository file structure alongside commit messages that document the progression of backend scripting, database integration, and frontend updates completed during the week.

WEEK 6: DATABASE INTEGRATION AND CRUD OPERATIONS

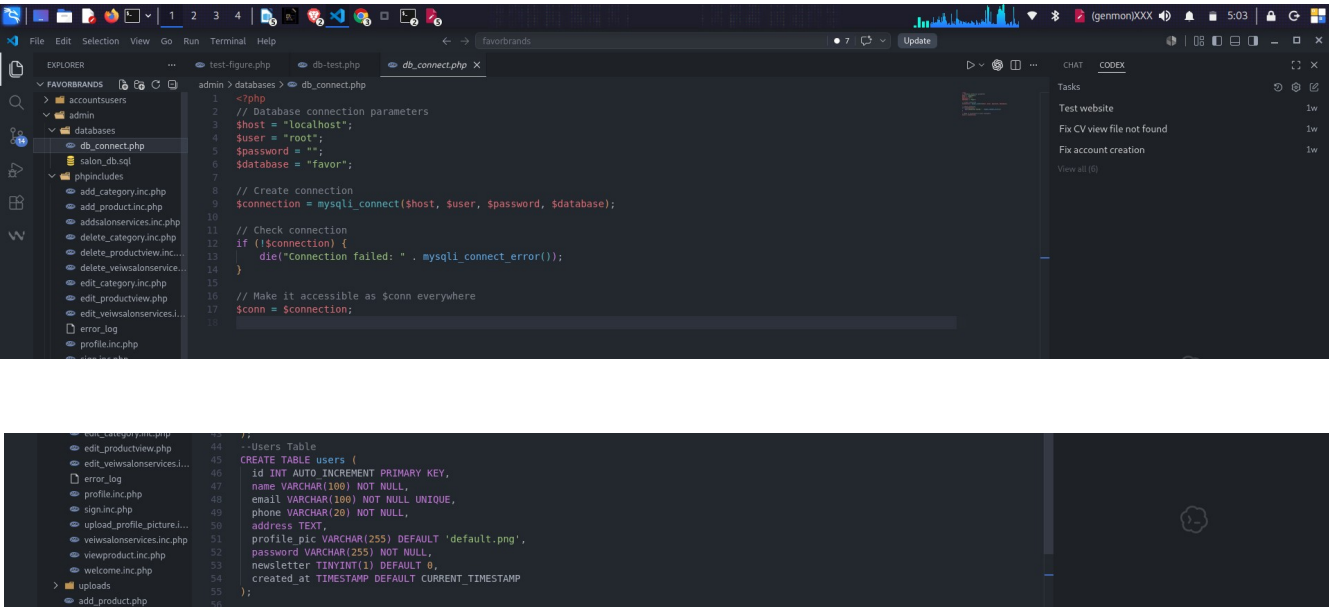


Figure 6.1: Database Structure & Connection

This figure displays the MySQL database structure and the successful PHP database connection configuration.

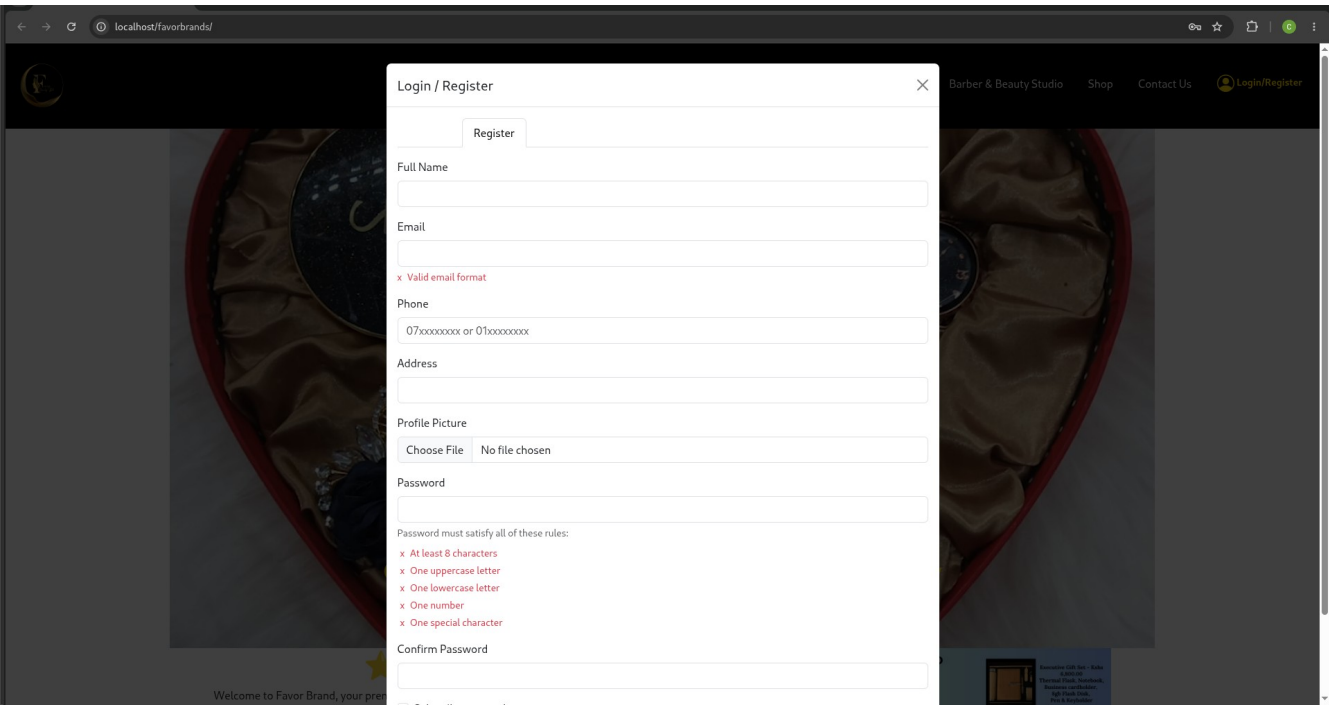


Figure 6.2: Clients Registration Form

This figure illustrates the clients registration interface with client-side validation and input fields for full name, email, and course selection.

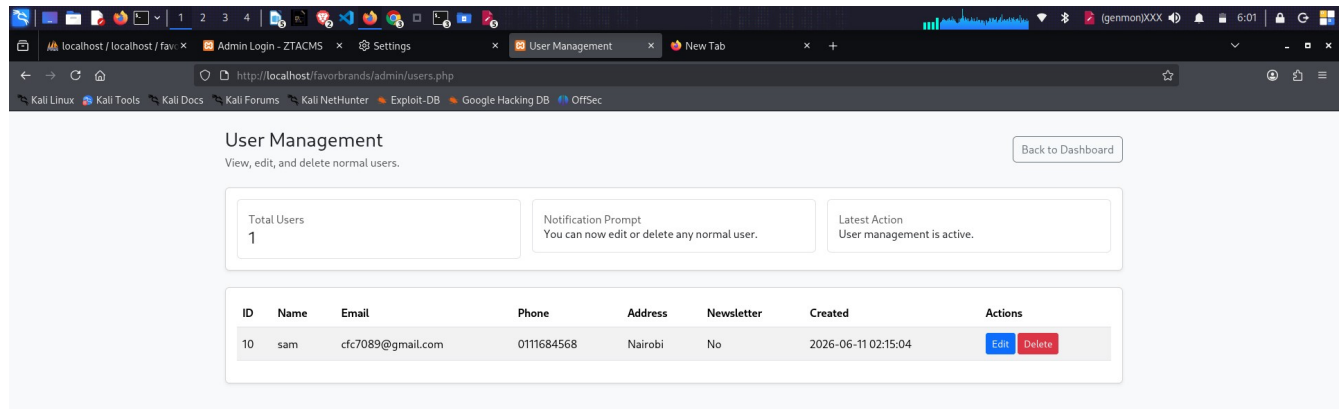


Figure 6.3: Clients Record Display

This figure shows the main dashboard displaying all clients records in a formatted table with edit and delete action buttons.

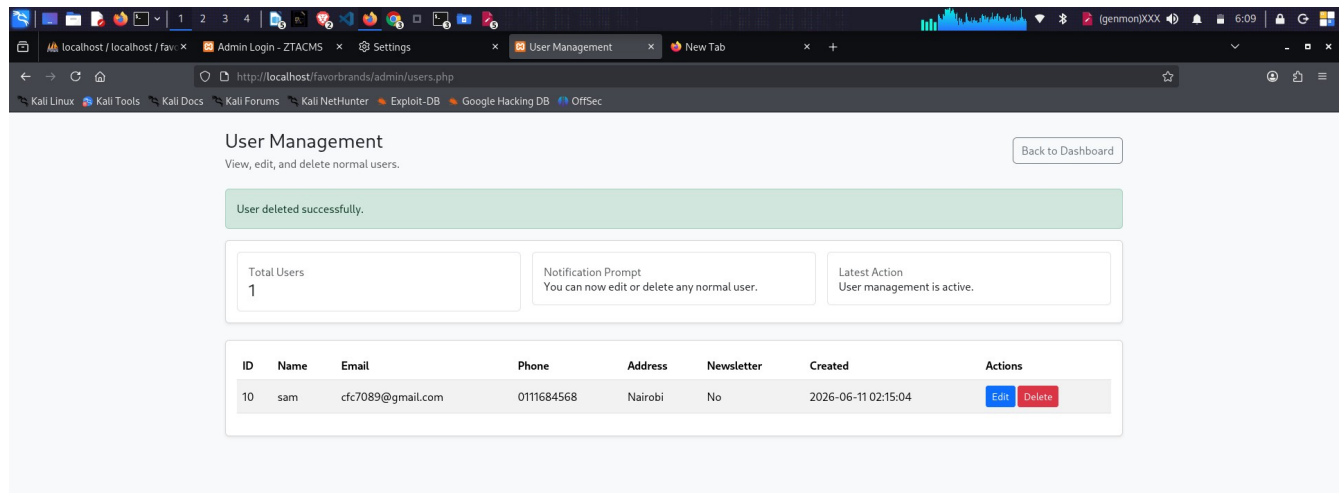


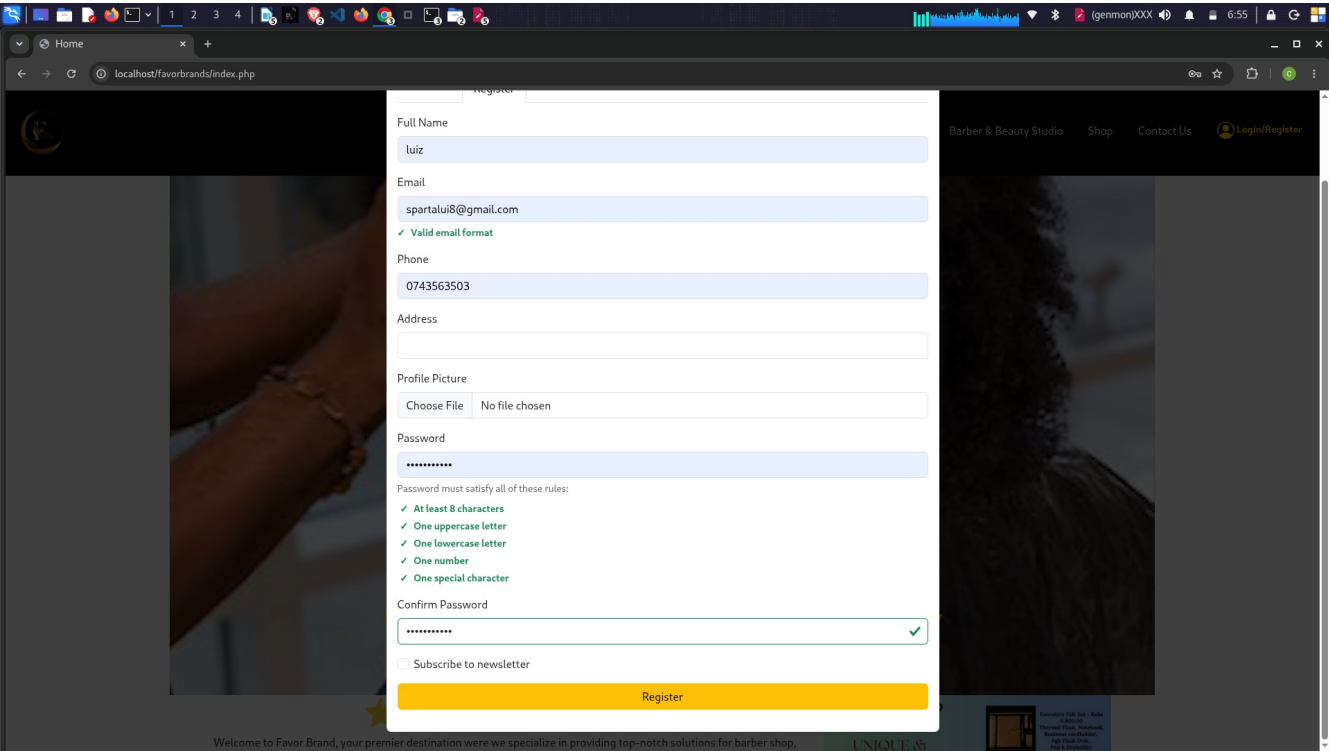
Figure 6.4: Clients Delete Confirmation

This figure captures the JavaScript confirmation dialog for clients record deletion.

Reflection

This week 6 successfully implemented complete CRUD operations for a clients management system. I learned the importance of prepared statements to prevent SQL injection and the value of both client and server-side validation. The practical experience of building a full data management interface has solidified my understanding of database integration in web applications.

WEEK 7: USER AUTHENTICATION AND SESSION MANAGEMENT



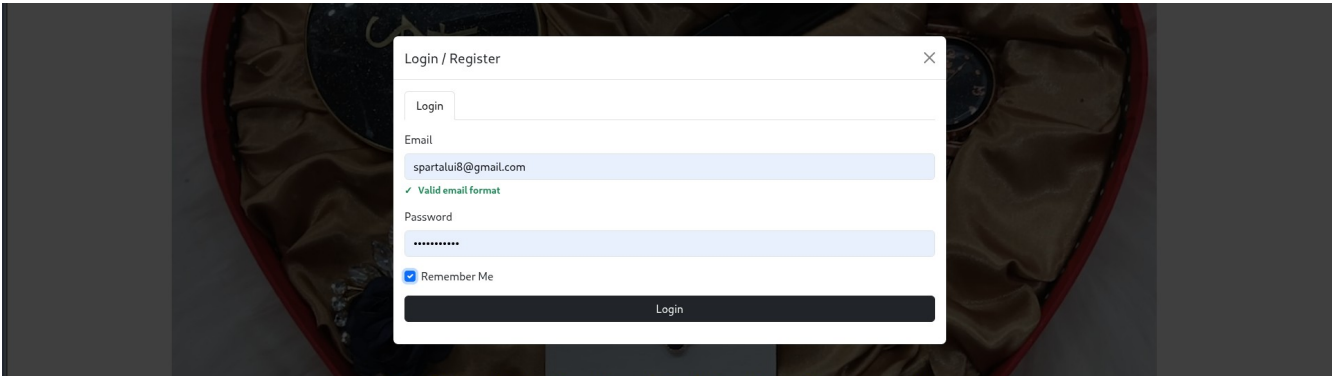
The screenshot shows a web browser window with the URL `localhost/favorbrands/index.php`. The page features a registration form with the following fields and elements:

- Full Name:** Input field containing "luiz".
- Email:** Input field containing "spartalui8@gmail.com" with a green checkmark and the text "Valid email format".
- Phone:** Input field containing "0743563503".
- Address:** Empty input field.
- Profile Picture:** A button labeled "Choose File" and the text "No file chosen".
- Password:** Input field with masked characters "*****". Below it, a list of password requirements is shown with green checkmarks:
 - At least 8 characters
 - One uppercase letter
 - One lowercase letter
 - One number
 - One special character
- Confirm Password:** Input field with masked characters "*****" and a green checkmark.
- Subscribe to newsletter:** An unchecked checkbox.
- Register:** A prominent yellow button.

The background of the page shows a blurred image of a person's arm and a watch.

Figure 7.1: User Registration Interface

This figure displays the complete registration form with password strength meter and validation feedback for creating new user accounts.



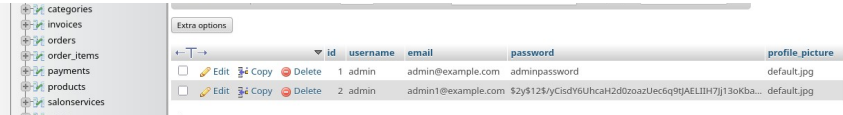
The screenshot shows a modal window titled "Login / Register" with a close button (X). The modal contains the following elements:

- Login / Register:** A tabbed interface with "Login" selected.
- Email:** Input field containing "spartalui8@gmail.com" with a green checkmark and the text "Valid email format".
- Password:** Input field with masked characters "*****".
- Remember Me:** A checked checkbox.
- Login:** A dark button.

The background of the page shows a blurred image of a person's arm and a watch.

Figure 7.2: Login Form and Session Management

This figure shows the secure login interface with "Remember Me" checkbox, error handling, and successful session creation after authentication.



The screenshot shows a database table with the following data:

	id	username	email	password	profile_picture
☐ Edit <input type="copy"/> <input type="delete"/>	1	admin	admin@example.com	adminpassword	default.jpg
☐ Edit <input type="copy"/> <input type="delete"/>	2	admin	admin1@example.com	\$2y\$12\$yCisdY6UhcH2d0zoazUec6g9jAELIH7j13oKba...	default.jpg

Figure 7.3: Password Hashing and Database Storage

This figure demonstrates bcrypt password hashing implementation and the secure storage of user credentials in the database.

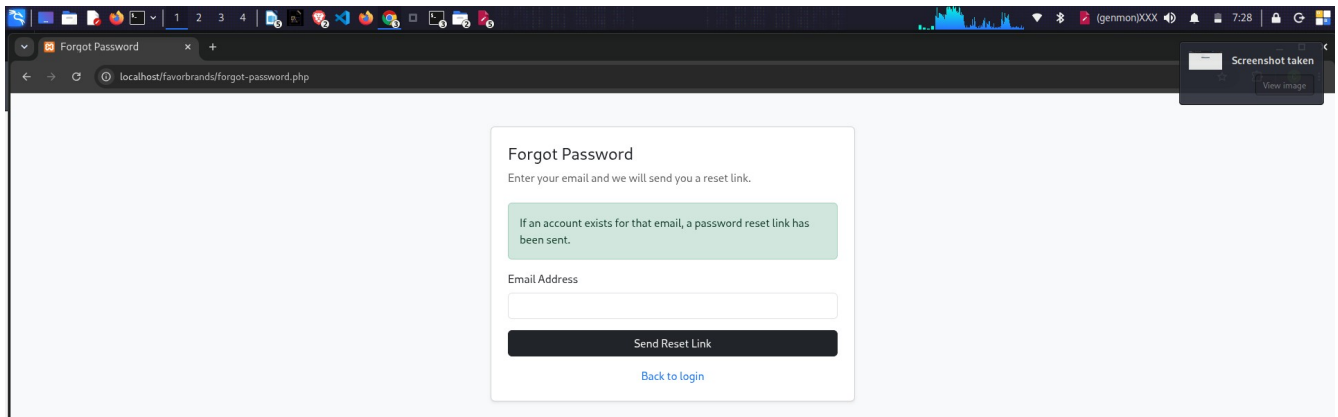


Figure 7.4: Reset-Password-Flow

This figure captures the password reset functionality including token generation, email verification, and secure password update process.

Reflection

This week I implemented a comprehensive authentication system with multiple security layers. The most valuable lesson was understanding password hashing and session management for secure web applications. The role-based authorization system taught me how to control user access effectively. I now understand why security is critical in web development.